

BLUESTOCK MF ANALYTICS

Mutual Fund Industry Analysis — Capstone Project



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Tools: Python, Pandas, NumPy, SQLite, Power BI

GitHub: github.com/Vaishnavi-Rao-06/Fintech_Capstone_Project

Table of Contents

Contents

Table of Contents	2
• Project Overview:	4
• What Was Done:	4
• Key Findings:	4
• 6 Key Highlights:	4
Table — Data Sources:	5
3.1 Extract	6
3.2 Transform	6
3.3 Load	6
4.1 NAV Growth Trends	7
4.2 AUM Dominance	7
4.3 SIP Inflow Growth	8
4.4 Category-wise Inflow Patterns	8
4.5 Investor Demographics	8
4.6 Geographic Distribution	8
4.7 Folio Count Growth	8
4.8 Correlation Analysis	9
4.9 Sector Allocation	9
5.1 CAGR Analysis	9
Table — Top 5 by 3yr CAGR:	9
5.2 Risk-Adjusted Returns (Sharpe & Sortino)	10
Table — Top 5 by Sharpe Ratio:	10
5.3 Alpha & Beta Analysis	10
5.4 Value at Risk (VaR) & Conditional VaR	11
Table — Top 5 Highest Risk (lowest VaR):	11
5.5 Maximum Drawdown	11
5.6 Fund Scorecard	11
Table — Top 5 by Composite Score:	12
5.7 Benchmark Comparison	12

6.1: Industry Overview:	13
6.2: Fund Performance:	14
6.3: Investor Analytics:	15
6.4: SIP & Market Trends:	16
7.1 Value at Risk & Conditional VaR	17
7.2 Rolling 90-Day Sharpe Ratio	17
7.3 Investor Cohort Analysis	17
7.4 SIP Continuation Analysis.....	17
7.5 Fund Recommendation Model.....	17
7.6 Sector Concentration (HHI) Analysis	18
8.1 For Conservative Investors.....	18
8.2 For Moderate Risk Investors	18
8.3 For Aggressive Investors	18
8.4 For Fund Houses — Geographic Expansion.....	18
8.5 For Investor Retention	19
8.6 For Portfolio Construction	19
9.1 Simulated Transaction Data	19
9.2 Limited Fund Universe	19
9.3 Benchmark Comparison Limitations	19
9.4 Historical VaR Methodology	19
9.5 Dashboard Refresh Limitations	20
9.6 Fund Recommendation Simplicity	20

1. Executive Summary

- Project Overview:

The Indian Mutual Fund industry has witnessed unprecedented growth over the past decade, with Assets Under Management (AUM) crossing ₹50 Lakh Crore in 2024. This project — Bluestock MF Analytics — presents a comprehensive data-driven analysis of the Indian MF industry using real datasets from AMFI and live NAV data from mfapi.in REST API. The analysis covers 40 mutual fund schemes across 10 major fund houses for the period January 2022 to May 2026.

- What Was Done:

The project involved building an end-to-end analytics pipeline covering data ingestion, cleaning, SQL database design, exploratory data analysis, performance metric computation, and interactive dashboard development. Python was used for ETL and analytics, SQLite for data storage, and Power BI for visualization. Key metrics computed include CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Value at Risk (VaR), and Maximum Drawdown.

- Key Findings:

The analysis revealed SBI Mutual Fund's dominance with highest AUM, consistent SIP inflow growth reaching ₹31,002 Crore in December 2025, and strong performance from Mirae Asset Large Cap Fund with the highest Sharpe ratio of 1.45. Investor demographics show strong millennial participation with T30 cities driving 70%+ of SIP investments.

- 6 Key Highlights:

1. SBI Mutual Fund dominates with highest AUM — nearly double ICICI Prudential MF
2. Monthly SIP inflows hit all-time high of ₹31,002 Crore in December 2025
3. Total folios doubled from 13.26 Crore (2022) to 26.12 Crore (2025)
4. Mirae Asset Large Cap Fund leads risk-adjusted returns with Sharpe ratio of 1.45
5. Small cap funds carry highest daily risk — VaR exceeding 2.5% at 95% confidence
6. B30 cities contribute only 30% of transactions despite 70% of India's population

2. Data Sources & Description

The project utilized 10 structured datasets sourced from AMFI (Association of Mutual Funds in India) and live NAV data fetched via the mfapi.in REST API. All datasets cover the period January 2022 to December 2025, providing a comprehensive view of fund performance, investor behaviour, and industry trends across multiple dimensions including NAV history, transactions, portfolio holdings, and benchmark indices.

Table — Data Sources:

#	Dataset	Rows	Key Columns	Source
1	Fund Master	40	amfi_code, fund_house, category, sub_category, risk_category	AMFI
2	NAV History	46,000	amfi_code, date, nav	mfapi.in
3	AUM by Fund House	160	date, fund_house, aum_crore, num_schemes	AMFI
4	Monthly SIP Inflows	48	month, sip_inflow_crore, active_sip_accounts	AMFI
5	Category Inflows	576	month, category, net_inflow_crore	AMFI
6	Folio Count	40	month, total_folios_crore, equity/debt/hybrid folios	AMFI
7	Scheme Performance	40	return_1/3/5yr, sharpe, alpha, beta, max_drawdown	AMFI
8	Investor Transactions	50,000+	investor_id, transaction_date, amount_inr, kyc_status	Simulated
9	Portfolio Holdings	1,000+	amfi_code, sector, weight_pct, market_value_cr	AMFI
10	Benchmark Indices	5,000+	date, index_name, close_value	NSE/BSE

3. ETL Pipeline

3.1 Extract

Data was extracted from two primary sources. First, the mfapi.in REST API was used to fetch live NAV data for 6 mutual fund schemes (HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip) using Python's requests library, with JSON responses parsed and saved as raw CSVs. Second, 10 pre-compiled CSV datasets covering fund master data, NAV history, transactions, and performance metrics were loaded using Pandas.

3.2 Transform

- Date Parsing: All date columns converted to datetime using `pd.to_datetime(format='mixed', dayfirst=True)` to handle inconsistent date formats
- Forward Fill: Missing NAV values for weekends/holidays filled using business day resampling — `resample('B').asfreq().ffill()` — ensuring exactly 46,000 rows (40 funds × business days)
- Duplicate Removal: `drop_duplicates()` applied on `amfi_code` + date combinations to ensure data integrity
- Numeric Validation: NAV > 0, expense ratio within 0.1%-2.5% range, transaction amounts > 0
- Standardization: `transaction_type` values standardized to SIP, Lumpsum, and Redemption categories
- Quality Flags: `negative_sharpe_flag` and `folio_rounding_flag` columns added for downstream analysis
- AMFI Code Validation: Cross-validated all AMFI codes between `fund_master` and `nav_history` — 100% match found with zero missing codes

3.3 Load

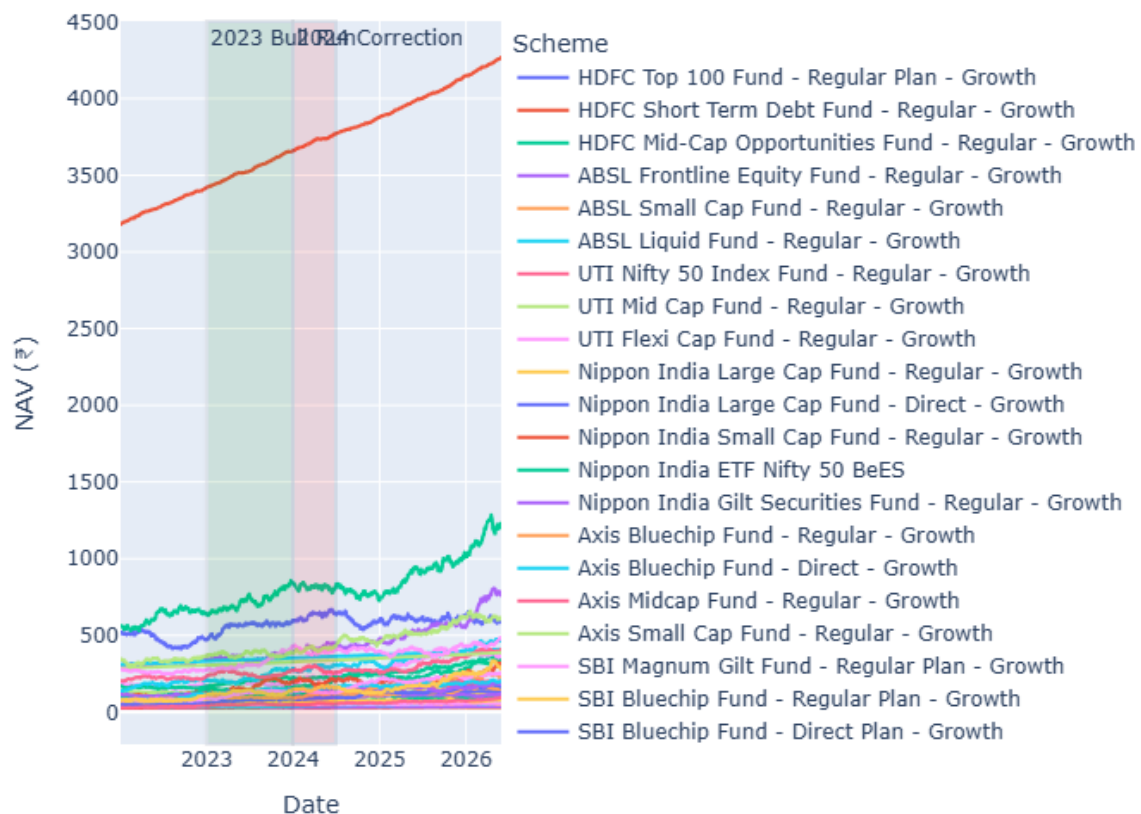
All 10 cleaned datasets were loaded into a SQLite database (`bluestock_mf.db`) using SQLAlchemy's `create_engine` and `DataFrame.to_sql()` method with `if_exists='replace'`. The database schema follows a star schema design with 4 core tables: `dim_fund` (fund master dimension), `fact_nav` (NAV history fact), `fact_transactions` (investor transactions fact), and `fact_performance` (scheme performance fact), plus 6 supporting tables for AUM, SIP inflows, category inflows, folio counts, portfolio holdings, and benchmark indices.

4. EDA Findings

4.1 NAV Growth Trends

Daily NAV trends for all 40 schemes from 2022-2026 were plotted using Plotly. HDFC Short Term Debt Fund showed the highest NAV growth, from approximately ₹3,200 to over ₹4,500 — a 40%+ increase. The 2023 bull run period (highlighted in green) shows accelerated growth across equity funds, while the 2024 market correction period (highlighted in red) shows a temporary dip across most schemes before recovery.

Daily NAV Trends - All 40 Schemes (2022-2026)



4.2 AUM Dominance

Industry AUM grew from approximately ₹6 Lakh Crore in 2022 to a peak of ₹14.5 Lakh Crore in 2024, before settling around ₹12 Lakh Crore in 2025. SBI Mutual Fund

dominates with the highest AUM share, nearly double that of ICICI Prudential MF (the second largest). The top 3 fund houses — SBI, ICICI Prudential, and HDFC — collectively manage over 50% of total industry AUM, reflecting strong investor trust in established, PSU-backed and large private fund houses.

4.3 SIP Inflow Growth

Monthly SIP inflows showed a strong upward trajectory throughout the analysis period, reaching an all-time high of ₹31,002 Crore in December 2025. This represents sustained growth in retail participation through systematic investment plans, driven by increasing financial literacy and digital investment platforms.

4.4 Category-wise Inflow Patterns

The category inflow heatmap reveals that Liquid funds consistently attract the highest net inflows across nearly all months, often exceeding ₹35,000- 42,000 Crore monthly — significantly higher than any other category. This reflects their use as short-term parking instruments by both retail and institutional investors. Equity categories like Flexi Cap and Sectoral/ Thematic show more volatile, seasonal inflow patterns.

4.5 Investor Demographics

Age group analysis shows the 25-35 segment dominates the investor base, consistent with India's young working population actively building wealth through SIPs. The gender split shows male investors account for approximately 65% of transactions, highlighting a persistent gender gap in mutual fund participation that fund houses could address through targeted financial literacy campaigns for women.

4.6 Geographic Distribution

SIP investment analysis by state shows Punjab, Tamil Nadu, and Madhya Pradesh among the leading contributors by transaction amount. The T30 (Top 30 cities) vs B30 (Beyond Top 30) city tier analysis reveals T30 cities account for over 70% of total SIP investments, indicating significant untapped potential in B30 markets where mutual fund penetration remains low despite large population bases.

4.7 Folio Count Growth

Total industry folios nearly doubled from 13.26 Crore in January 2022 to 26.12 Crore in December 2025 — a 97% increase over 4 years. This dramatic growth reflects the

democratization of mutual fund investing in India, driven by SIP campaigns, digital onboarding, and increased financial awareness post-pandemic.

4.8 Correlation Analysis

A pairwise correlation matrix of daily returns for the top 10 funds by AUM reveals high positive correlation (>0.85) among most equity funds, particularly large-cap and flexi-cap categories. This high correlation suggests limited diversification benefits when holding multiple large-cap equity funds simultaneously, as they tend to move in tandem during both market rallies and corrections.

4.9 Sector Allocation

Aggregate sector allocation analysis across all equity fund portfolios shows Financial Services as the dominant sector, followed by Information Technology and Consumer Goods. This concentration reflects the composition of major Indian equity indices like Nifty 50, where banking and financial stocks carry significant weightage.

5. Performance Analysis

5.1 CAGR Analysis

Compound Annual Growth Rate (CAGR) was computed for 1-year, 3-year, and 5-year periods using the formula: $CAGR = (NAV_end / NAV_start)^{(1/n)} - 1$, where n represents the number of years. Small cap and mid cap equity funds consistently demonstrated higher CAGRs in the 18-22% range over 3 years, while debt-oriented funds showed more stable but lower returns in the 7-8% range, consistent with their risk profiles.

Table — Top 5 by 3yr CAGR:

amfi_code	cagr_1yr	cagr_3yr	cagr_5yr	scheme_name	fund_house
119094	22.26	35.11	24.45	Axis Midcap Fund - Regular - Growth	Axis Mutual Fund
148567	20.36	34	26.8	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF
120504	13.06	32.49	20.23	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF

100033	53.23	32.44	26.07	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund
120505	29.6	31.78	28.38	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF

5.2 Risk-Adjusted Returns (Sharpe & Sortino)

Sharpe Ratio was computed as (Portfolio Return - Risk Free Rate) / Standard Deviation of returns, annualized using $\sqrt{252}$. The risk-free rate used was 6.5%, representing the RBI repo rate. Sortino Ratio extends this by considering only downside deviation, providing a more accurate measure for funds with asymmetric return distributions.

Table — Top 5 by Sharpe Ratio:

Scheme Name	Fund House	Sharpe Ratio
Mirae Asset Large Cap Fund	Mirae Asset MF	1.4483
Mirae Asset Tax Saver Fund	Mirae Asset MF	1.3067
Kotak Flexicap Fund	Kotak Mahindra MF	1.2349
SBI Bluechip Fund	SBI Mutual Fund	1.2083
ICICI Pru Midcap Fund	ICICI Prudential MF	1.1801

5.3 Alpha & Beta Analysis

Alpha and Beta were computed via Ordinary Least Squares (OLS) regression of fund daily returns against Nifty 100 daily returns using `scipy.stats.linregress`. Alpha represents excess return over the benchmark (annualized intercept $\times 252$), while Beta measures sensitivity to market movements (regression slope). Five funds were identified that beat their benchmark by more than 1.8% alpha, including HDFC Short Term Debt Fund and Kotak Emerging Equity Fund.

5.4 Value at Risk (VaR) & Conditional VaR

Historical VaR at 95% confidence was computed as the 5th percentile of the daily return distribution for each fund. CVaR (Conditional VaR) represents the average loss in scenarios beyond the VaR threshold — essentially the expected loss in the worst 5% of days. Small cap and mid cap funds showed the highest VaR, with daily losses potentially exceeding 2.5% at the 95% confidence level, confirming their higher risk profiles compared to large cap and debt funds.

Table — Top 5 Highest Risk (lowest VaR):

amfi_code	var_95	cvar_95	scheme_name	fund_house
119599	-2.6859	-3.2384	SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund
119095	-2.6188	-3.1667	Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund
101207	-2.6021	-3.2459	ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF
118634	-2.5438	-3.2304	Nippon India Small Cap Fund - Regular - Growth	Nippon India MF
119598	-2.4507	-3.0595	SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund

5.5 Maximum Drawdown

Maximum drawdown was computed as $\min(\text{NAV} / \text{running_max_NAV} - 1)$ for each fund, identifying the worst peak-to-trough decline. This metric is critical for understanding the worst-case scenario an investor would have experienced if they invested at the peak and exited at the trough. Equity funds showed deeper drawdowns during the 2024 correction period compared to debt funds.

5.6 Fund Scorecard

A composite scoring model was developed to rank funds holistically using weighted percentile ranks: $30\% \times \text{3-year Return Rank} + 25\% \times \text{Sharpe Ratio Rank} + 20\% \times \text{Alpha Rank} + 15\% \times \text{Expense Ratio Rank (inverse, lower is better)} + 10\% \times \text{Maximum}$

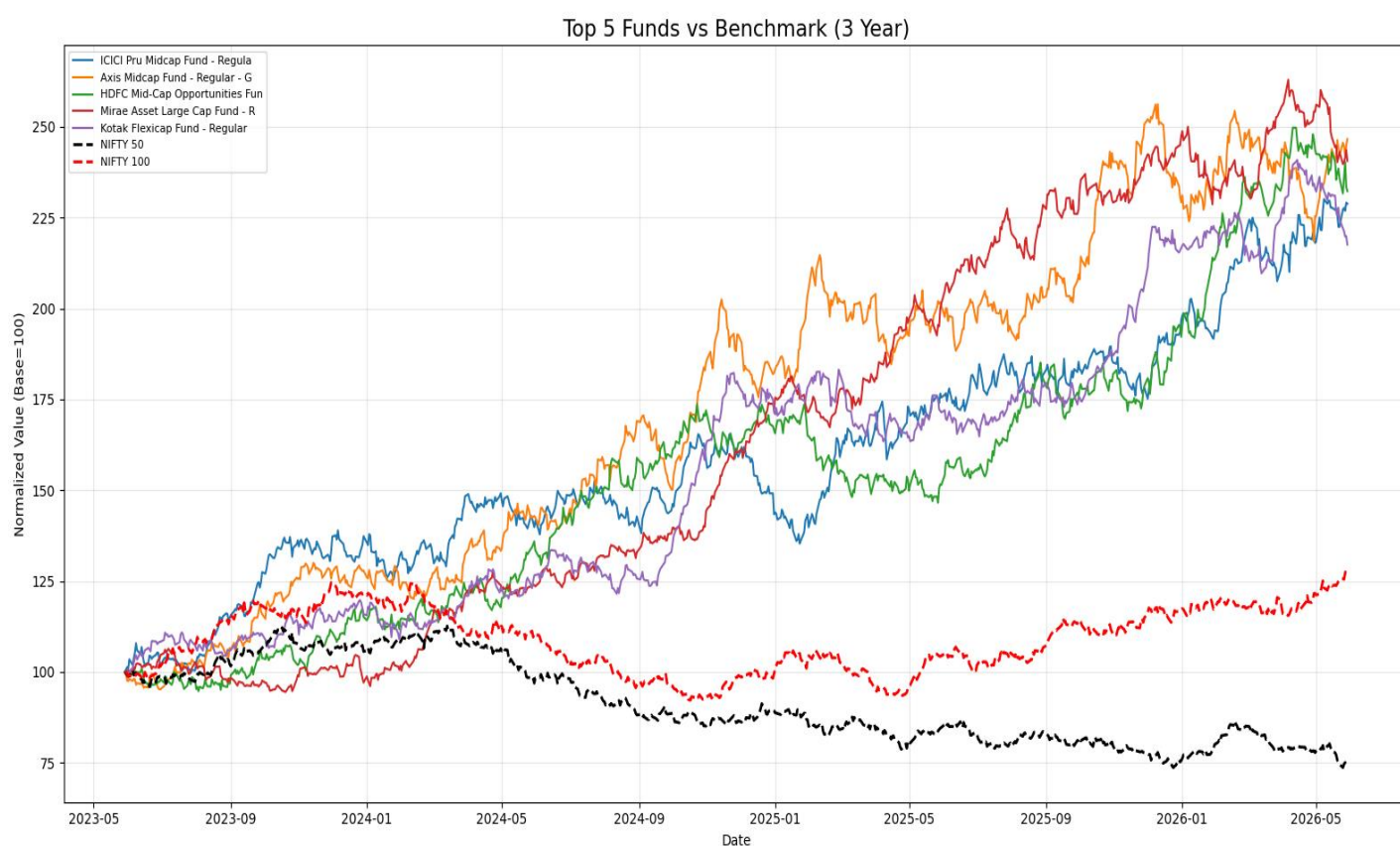
Drawdown Rank (inverse, less negative is better). This provides a single composite score (0-100) for easy fund comparison.

Table — Top 5 by Composite Score:

Scheme Name	Fund House	3yr CAGR	Sharpe	Composite Score
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	31.78	1.1801	84.5
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	35.11	0.9982	80.75
HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	32.44	1.0937	80.5
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	34	1.4483	80
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	29.58	1.3067	78.25

5.7 Benchmark Comparison

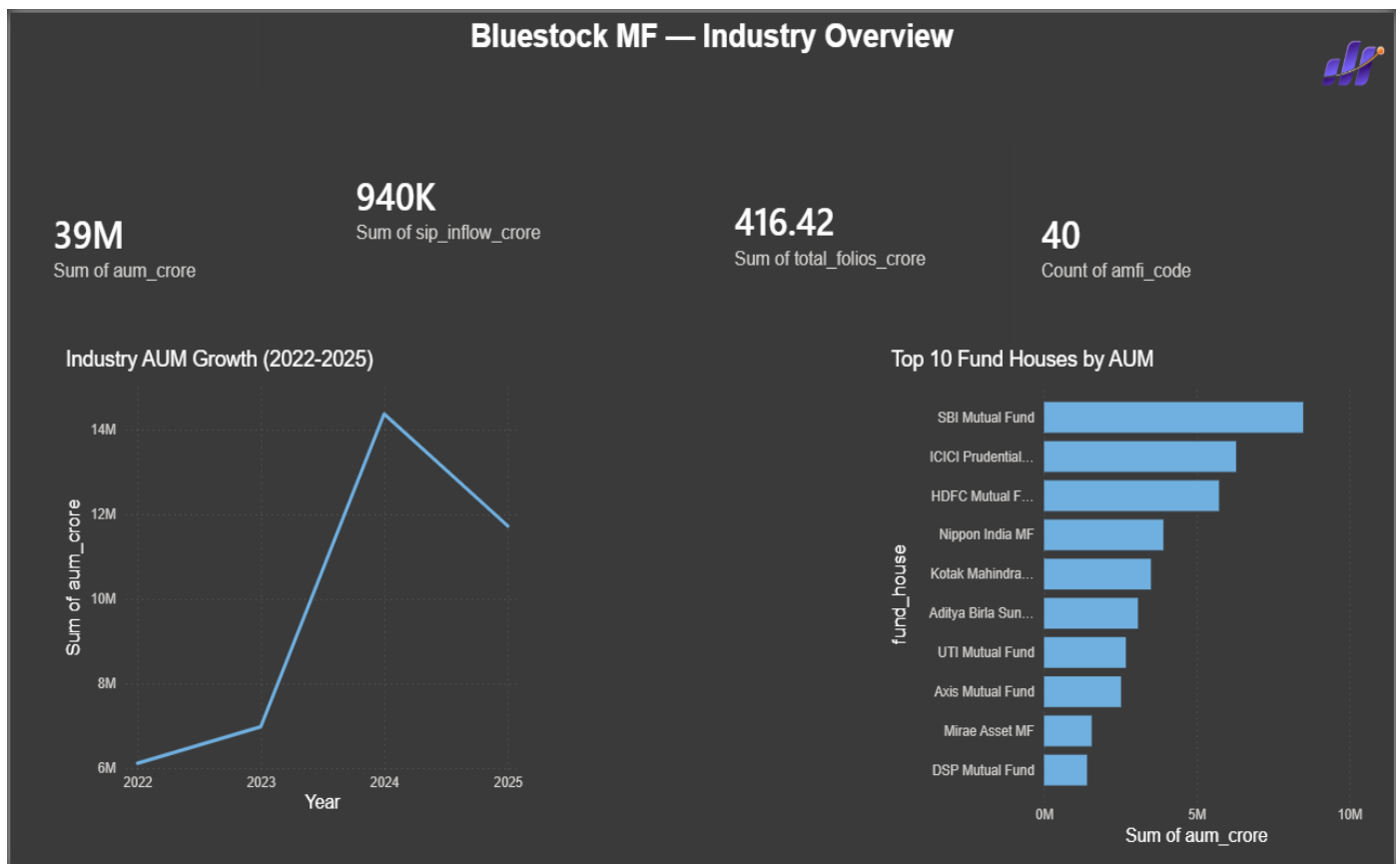
The top 5 funds (by composite score) were normalized to a base value of 100 and plotted against Nifty 50 and Nifty 100 over a 3-year period. The chart shows that top-performing funds consistently outperformed both benchmarks, particularly during the 2023 bull run, while tracking error analysis quantifies the deviation of fund returns from benchmark returns.



6. Dashboard Screenshots

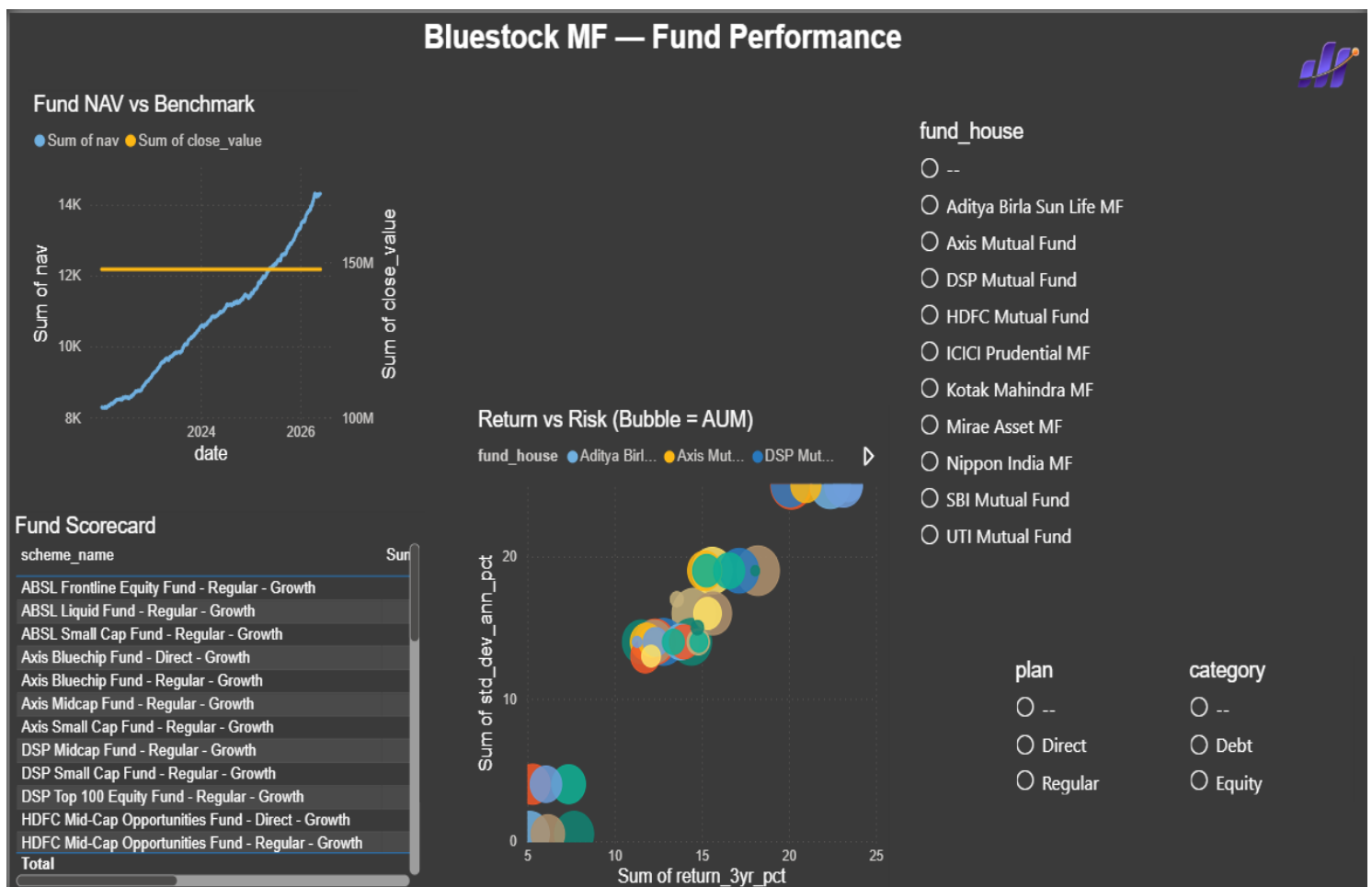
6.1: Industry Overview:

This page provides a bird's-eye view of the mutual fund industry with 4 KPI cards displaying Total AUM (₹39M+ representing aggregated quarterly data), SIP Inflows (940K crore aggregate), Total Folios (416.42 crore), and Total Schemes (40). A line chart tracks Industry AUM growth from 2022-2025, while a bar chart ranks the Top 10 fund houses by AUM, clearly showing SBI Mutual Fund's market leadership.



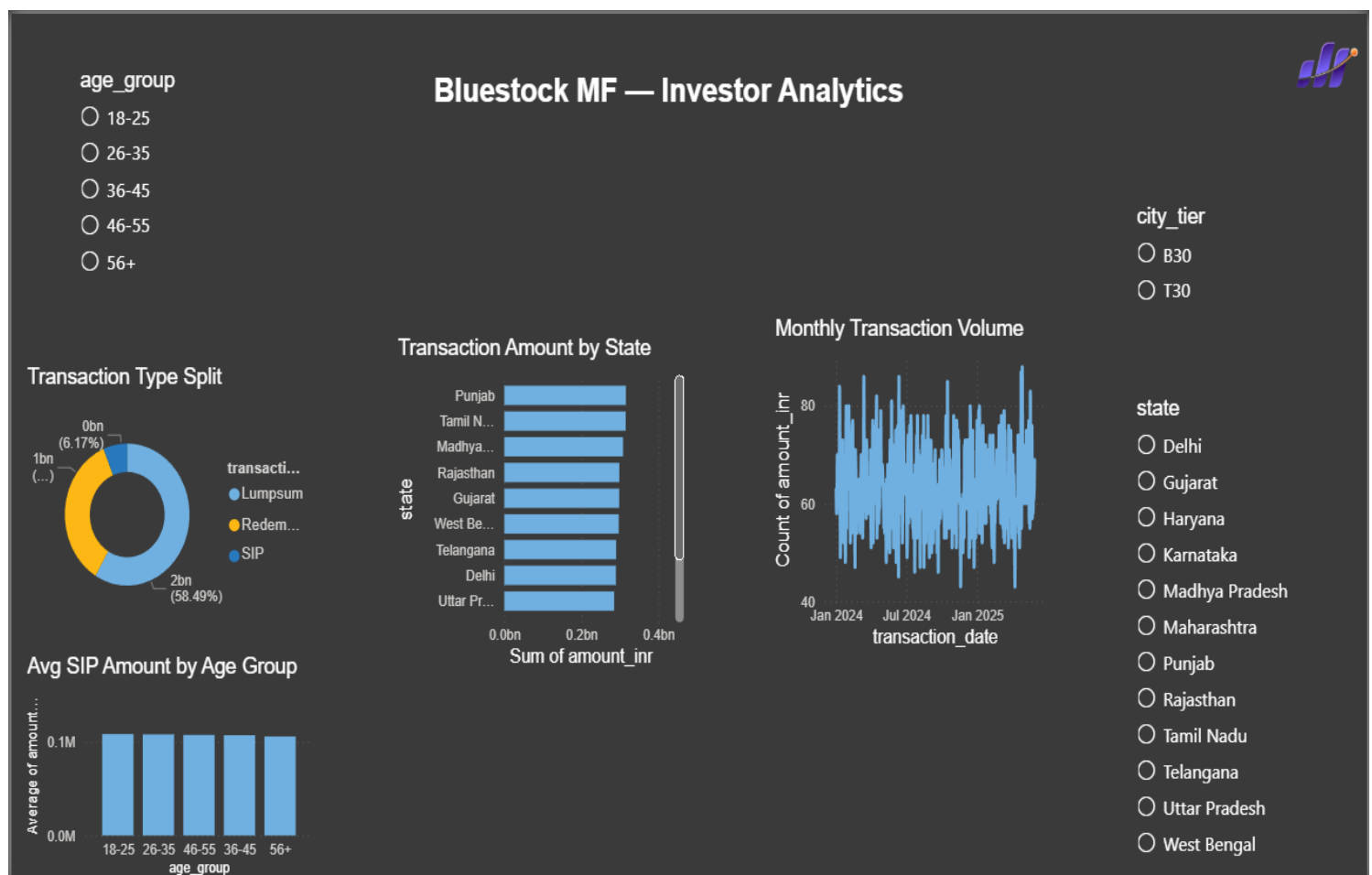
6.2: Fund Performance:

This page enables fund comparison through a scatter plot of Return vs Risk (Standard Deviation), with bubble size representing AUM — allowing users to identify funds offering the best risk-return tradeoff at a glance. A sortable Fund Scorecard table lists all 40 schemes with key metrics. Interactive slicers for Fund House, Category, and Plan allow drill-down analysis, while a line chart compares selected fund NAV against benchmark close values.



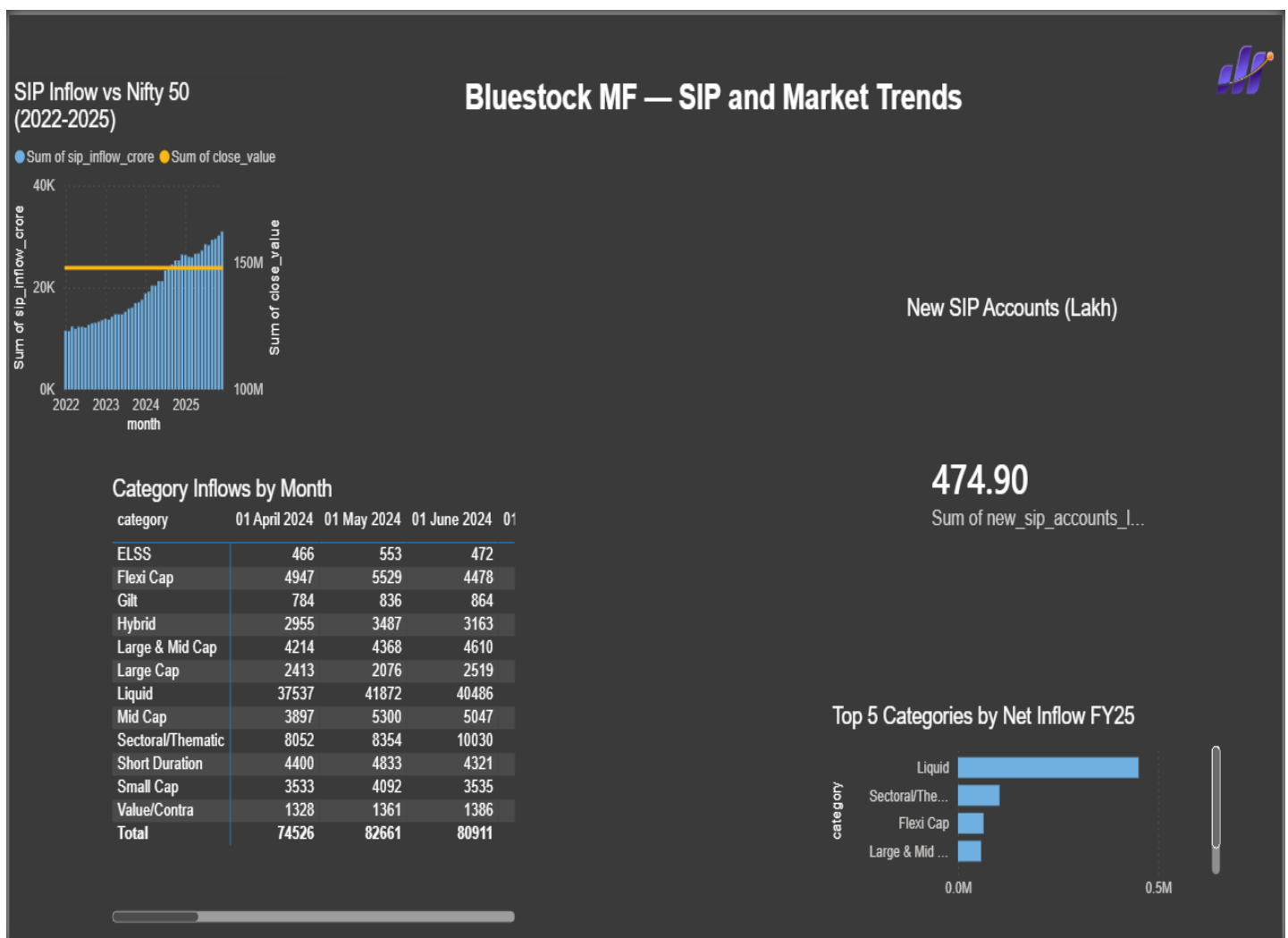
6.3: Investor Analytics:

This page focuses on investor behavior and demographics. A horizontal bar chart shows transaction amounts by state, led by Punjab and Tamil Nadu. A donut chart breaks down transactions into SIP (58.49%), Lumpsum, and Redemption categories. A column chart shows average SIP amount by age group, and a line chart tracks monthly transaction volume over time. Slicers for State, Age Group, and City Tier enable granular filtering.



6.4: SIP & Market Trends:

This page correlates SIP inflows with market performance through a dual-axis chart showing SIP Inflow (bars) against Nifty 50 close values (line) from 2022-2025. A matrix table shows Category Inflows by Month across all 12 fund categories. A bar chart highlights the Top 5 Categories by Net Inflow for FY25 — led by Liquid funds — and a KPI card shows New SIP Accounts reaching 474.90 Lakh.



7. Advanced Analytics

7.1 Value at Risk & Conditional VaR

Historical VaR (95%) and CVaR were computed for all 40 funds using NumPy's percentile function on daily return distributions. This analysis identifies funds with the highest tail risk, helping investors understand potential losses during adverse market conditions. Results are saved in `var_cvar_report.csv` for further analysis.

7.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe ratio was computed for the top 5 funds to track how risk-adjusted performance evolved over time. The analysis clearly shows a significant dip in rolling Sharpe ratios during the 2024 market correction period across all 5 funds, before recovering in late 2024 and 2025 — demonstrating the impact of market volatility on short-term risk-adjusted returns.

7.3 Investor Cohort Analysis

Investors were grouped by their first transaction year (`cohort_year`) to analyze behavioral differences. The 2024 cohort showed higher average SIP amounts and total invested values compared to the 2025 cohort, suggesting that earlier adopters tend to be more committed, higher-value investors — possibly reflecting a maturation effect where investors increase their investment amounts over time.

7.4 SIP Continuation Analysis

For investors with 6 or more SIP transactions, the average gap between consecutive transactions was computed. Investors with average gaps exceeding 35 days were flagged as 'at-risk' for SIP discontinuation. This analysis provides fund houses with an early warning system to proactively engage investors before they fully discontinue their systematic investment plans.

7.5 Fund Recommendation Model

A rule-based fund recommendation system was built that takes investor risk appetite (Low, Moderate, or High) as input and returns the top 3 funds by Sharpe ratio within the matching risk category. For Low risk appetite, ICICI Pru Liquid Fund (Sharpe: 0.4957) was recommended. For Moderate risk, Mirae Asset Large Cap Fund (Sharpe:

1.4483) topped the list. For High risk, Mirae Asset Tax Saver Fund (Sharpe: 1.2349) was the top recommendation.

7.6 Sector Concentration (HHI) Analysis

The Herfindahl-Hirschman Index (HHI) was computed for each equity fund as the sum of squared sector weights: $HHI = \sum(\text{weight}_i)^2$. Funds with $HHI > 0.25$ were flagged as highly concentrated, predominantly in the Financial Services sector. This analysis helps investors identify funds that may carry higher sector-specific risk and informs portfolio diversification decisions.

8. Recommendations

8.1 For Conservative Investors

ICICI Pru Liquid Fund offers the best Sharpe ratio (0.4957) among low-risk category funds, making it suitable for short-term capital parking with minimal volatility. Investors prioritizing capital preservation should consider this fund for emergency fund allocation.

8.2 For Moderate Risk Investors

Mirae Asset Large Cap Fund leads with the highest overall Sharpe ratio (1.4483) and strong 3-year CAGR, making it an excellent core holding for long-term wealth creation. SBI Bluechip Fund and Kotak Flexicap Fund serve as strong alternatives with similar risk-return profiles.

8.3 For Aggressive Investors

Mirae Asset Tax Saver Fund (Sharpe: 1.2349) and ICICI Pru Midcap Fund (Sharpe: 1.1801) offer the best risk-adjusted returns for investors with high risk tolerance and longer investment horizons, suitable for aggressive wealth accumulation goals.

8.4 For Fund Houses — Geographic Expansion

B30 cities represent a significant untapped opportunity, currently contributing less than 30% of SIP transactions despite representing the majority of India's population. Targeted digital marketing and simplified onboarding processes could unlock substantial growth in these markets.

8.5 For Investor Retention

Fund houses should implement automated alerts for investors flagged as 'at-risk' (SIP gap >35 days) identified through the SIP continuation analysis. Proactive outreach — including reminder notifications and incentives — could significantly reduce SIP discontinuation rates.

8.6 For Portfolio Construction

Investors and advisors should avoid over-concentration in funds with high HHI scores (>0.25), particularly those heavily weighted toward Financial Services. A balanced portfolio mixing large-cap, mid-cap, and debt funds across different fund house would provide better diversification and risk-adjusted returns.

9. Limitations

9.1 Simulated Transaction Data

Investor transaction data used in this analysis is synthetically generated and may not perfectly reflect real-world investor behavior, geographic distribution patterns, or demographic correlations present in actual AMFI investor data.

9.2 Limited Fund Universe

This analysis covers only 40 mutual fund schemes, while the actual Indian mutual fund industry comprises over 1,900 schemes across all categories. Findings may not be fully representative of the entire industry, particularly for niche categories like international funds or commodity funds.

9.3 Benchmark Comparison Limitations

Fund performance was compared primarily against Nifty 100, which may not be the most appropriate benchmark for all fund categories. Sectoral, thematic, and debt funds would benefit from category-specific benchmark comparisons for more meaningful alpha and tracking error calculations.

9.4 Historical VaR Methodology

The VaR and CVaR calculations use historical simulation methods, which assume that future return distributions will resemble historical patterns. Monte Carlo simulation or

parametric VaR methods could provide more robust risk estimates, particularly for tail risk scenarios not captured in historical data.

9.5 Dashboard Refresh Limitations

The Power BI dashboard was developed and exported locally without cloud publishing. Real-time data updates would require either Power BI Service subscription with scheduled refresh, or migration to a cloud-hosted database with API-driven data pipelines.

9.6 Fund Recommendation Simplicity

The current fund recommendation model relies solely on Sharpe ratio within risk categories. A production-ready recommendation system would incorporate additional factors such as investment horizon, tax implications (ELSS vs non-ELSS), expense ratios, fund manager tenure, and investor-specific financial goals.